

Todd Cullen

Tech Leader. AI Engineer. Operator.

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WORK EXPERIENCE

100x (AI Product Consulting)

August 2023 - Present

AI Engineering Consultant

Seattle, WA

- **Voice Assistants:** Built an emotive voice assistant via a custom WebRTC protocol backed by Llama3 70B LLM running on Groq and a FastAPI web server managing a dynamic LLM context with tool calls.
- **Vector Data Pipelines:** Assembled batch processing pipelines to create vector embeddings with a range of open source models including Instructor, Word2Vec, and BERT for use with pgvector, Milvus, and Pinecone.
- **Chatbot + RAG:** Created a full stack web application (React, PostgreSQL, FastAPI, GPT-4o) to serve a chatbot with tool access to a range of proprietary information provided via a custom RAG backend.
- **AI Testing Systems:** Designed and implemented observability, evaluation, and telemetry systems supporting the production rollout of LLM-backed products with confidence.
- **Open Source Maintainer:** Built and actively maintain a number of open source libraries for working with LLMs: MindMeld (an AI Prompt Optimization Library), TypedLLM (a Pydantic LLM interface), and TypedTemplate (an LLM prompt library).

Evidence-Based Practice Institute (Mental Health)

April 2020 - July 2023

Chief Technology Officer & Chief Information Security Officer

Seattle, WA

- Managed a **budget of \$5M+** and a **team of 35+** people across five departments with seven direct reports.
- Led the 0 to 1 creation of the **first chatbot to assess and treat suicidal ideation**. Tech Stack: OpenAI GPT-3 (later upgraded to GPT-4) + Langchain + Ragas.
- Converted rack-based server infrastructure to docker-based containers running on **AWS deployed by Terraform**. Reduced costs by 75%, **saving over \$500k per year**.
- Upgraded the software, infrastructure, and operational processes to align with security best practices and **HIPAA, HITECH and NIST** standards **without increasing costs**.

Remarkably (Real Estate Marketing Intelligence - Acquired by Repivotly)

Jan. 2019 – April 2020

Chief Technology Officer & Head of Product

Seattle, WA

- **Ran Product and Engineering** while setting records for the fastest growing startup in Pioneer Square Labs history. ARR: \$600k @ 6mo, **\$1.2M @ 9mo**.
- **Saved \$1M+** on average Seattle engineer compensation by running a hybrid Seattle/Remote team while still hitting tight product deadlines.
- Built the first version of the product, a metrics and forecasting system for multifamily real estate operations.
- Built an Airflow ETL system on Google Cloud to convert our client's noisy property management systems data (Yardi, Real Page, Appfolio) to a clean source of operations metrics.

ThoughtLeadr (Digital Advertising - Acquired by Chaotic Moon)

May 2011 – December 2018

Founder & Chief Technology Officer

San Francisco, CA & Austin, TX

- Managed **10B+ requests per month** with peak load handling **over a 1 billion requests in a day**.
- **Beat Google's Ad Server (DFP)** by ~30% in time to first view and a 10%+ better CTR.
- Launched in 2011 to serve native ads for social sites including **Reddit, Imgur, 9gag**, and many more.
- Built a cutting edge ad server on **Haskell and ReactJS**.

ReignDesign (Mobile App Studio - Acquired Apply Digital)

Feb. 2009 – March 2011

Founder & Principal Engineer

Shanghai, China

- Developed the **first iOS app for the enterprise** for Standard Chartered Bank in Singapore.
- Sold the company's **first \$1M dollars in sales** to FrogDesign & Nortel.
- Pushed the organization into mobile development for **iOS and Android in 2009**.

- Designed an education workshop to convert developers from C++ to Objective-C to meet our client's growing mobile development needs.

EDUCATION

Wake Forest University

BA, Computational Biology

- Program focus: Genomics & Proteomics
- WAKE Radio: Station Manager
- Ultimate Frisbee: Captain

May, 2005

Winston-Salem, NC

Wake Forest University

Summer MBA Intensive Program

June, 2004

Winston-Salem, NC

AWARDS

Winner

Man vs Bear 10k

October, 2017

Destruction Bay, Yukon

Grant Recipient

Kauffman Foundation

June, 2005

Winston-Salem, NC

Eagle Scout

Boy Scouts of America

Long Time Ago

Galaxy Far Far Away

SKILLS

Business: Technical Leadership, Mentorship, Contract Negotiation, M&A, International Business, Digital Strategy, Digital Advertising Strategy, HIPAA/HITECH Compliance, Scrum, Agile
Software: Jira, Confluence, Bitbucket, Github, Asana, Figma, Aha!, Productboard, Trello
Programming: Python, Typescript, Javascript, Ruby, Haskell, Java, C#, SQL, Objective-C, Swift
LLMs: OpenAI, Llama, Mistral, Cohere, Perplexity, Groq, AnyScale, Eleven Labs, Humane
Databases: PostgreSQL, MySQL, Redis, Milvus, pgvector, Pinecone, Amazon RDS, MongoDB, Cassandra, Neo4J, Snowflake
Frontend Frameworks: React, React Native, Vue.js, Tailwind, Bulma, Bootstrap
Web Frameworks: Ruby on Rails, Django, FastAPI, NextJS
AI Frameworks: LangChain, Instructor, PyTorch, Ragas,
Infrastructure: AWS, Google Cloud, Azure, Terraform, CloudFormation, Render, Heroku